

# Fundamentos del aprendizaje automático

(Machine learning)

Joaquín Luque

# Contexto y fuentes

- Diversos proyectos de investigación en Machine Learning
  - Clasificación de sonidos, predicción en series temporales, ...
- CS 189/289A. Introduction to Machine Learning (2018-2019)
  - Universidad de Berkeley. Benjamin Recht, Moritz Hardt y Stella Yu
- Machine Learning Crash Course 2019
  - Universidad de Génova, MIT. Lorenzo Rosasco
- Catedra Indra “Sociedad Digital”

# Enfoque y audiencia

- Enfoque:
  - Introducción a la disciplina; o
  - Fundamentación de conocimientos previos
- No adecuado para:
  - Expertos en ML
  - Interesados en programación y aplicaciones
- Conocimientos previos aconsejables:
  - Álgebra matricial
  - Cálculo integral
  - Nociones de programación (Python u otros)
- Formalismos matemáticos:
  - Los “imprescindibles” para entender conceptos
  - Detalles formales opcionales (se facilitarán pero no se desarrollarán)

# Contenido

1. Introducción
2. Regresión
  - a) Regresión univariable
  - b) Regresión multivariable
3. Clasificación
  - a) Regresión logística
  - b) Máquinas de vectores soporte (SVM)
    - Forma dual de la optimización (regresión y SVM)
  - c) Funciones Kernel
  - d) Clasificación multiclase
4. Segmentación
5. Reducción de dimensionalidad
6. Deep learning (introducción)

Duración estimada: 10 semanas (15 horas)

# Referencias

- The Elements of Statistical Learning (Springer). Trevor Hastie et al.
  - <https://web.stanford.edu/~hastie/Papers/ESLII.pdf>
- Deep Learning (MIT Press). Ian Goodfellow et al.
  - <http://www.deeplearningbook.org/>
- Machine Learning
  - Coursera, Universidad de Stanford. Andrew Ng
  - <https://es.coursera.org/learn/machine-learning>
- Deep Learning Specialization
  - Coursera, Deeplearning.ai. Andrew Ng
  - [https://www.youtube.com/playlist?list=PLkDaE6sCZn6Ec-XTbcX1uRg2\\_u4xOEky0](https://www.youtube.com/playlist?list=PLkDaE6sCZn6Ec-XTbcX1uRg2_u4xOEky0) (y siguientes)

# Fundamentos del aprendizaje automático

## (Machine Learning)

Joaquín Luque Rodríguez

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### ● Presentaciones

- [Presentación del curso](#)
- [Introducción](#)
- [Regresión univariable](#)
- Regresión multivariable
- Clasificación
- Optimización dual
- Kernels
- Optimización multiclase
- Segmentación
- Reducción de dimensionalidad
- Deep Learning

### ● Ejercicios

- [Ejercicio 1. Preliminares de Python](#)  
[Datos](#)  
[Solución.](#)
- [Ejercicio 2. Regresión lineal univariable: Ecuación Normal.](#)  
[Solución.](#)
- Ejercicio 3. Regresión lineal univariable: Gradiente Descendiente.  
Solución.
- Ejercicio 4. Regresión polinómica univariable.

# Ejercicios

1. Preliminares de Python.
2. Regresión lineal univariable: Ecuación Normal.
3. Regresión lineal univariable: Gradiente Descendiente.
4. Regresión polinómica univariable.
5. Regresión multivariable.
6. Clasificación mediante regresión logística.
7. Clasificación mediante máquinas de vectores soporte (kernels).
8. Segmentación.
9. Reducción de dimensionalidad (PCA).
10. Redes neuronales.





Products

Why Anaconda?

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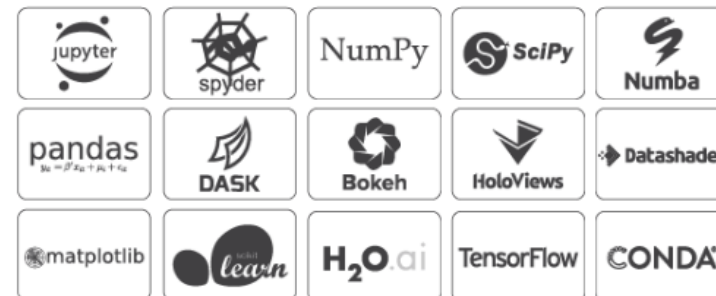
# Anaconda Distribution

The World's Most Popular Python/R Data Science Platform

Download

The open-source Anaconda Distribution is the easiest way to perform Python/R data science and machine learning on Linux, Windows, and Mac OS X. With over 15 million users worldwide, it is the industry standard for developing, testing, and training on a single machine, enabling *individual data scientists* to:

- Quickly download 1,500+ Python/R data science packages
- Manage libraries, dependencies, and environments with Conda
- Develop and train machine learning and deep learning models with scikit-learn, TensorFlow, and Theano
- Analyze data with scalability and performance with Dask, NumPy, pandas, and Numba
- Visualize results with Matplotlib, Bokeh, Datashader, and Holoviews



Windows



macOS



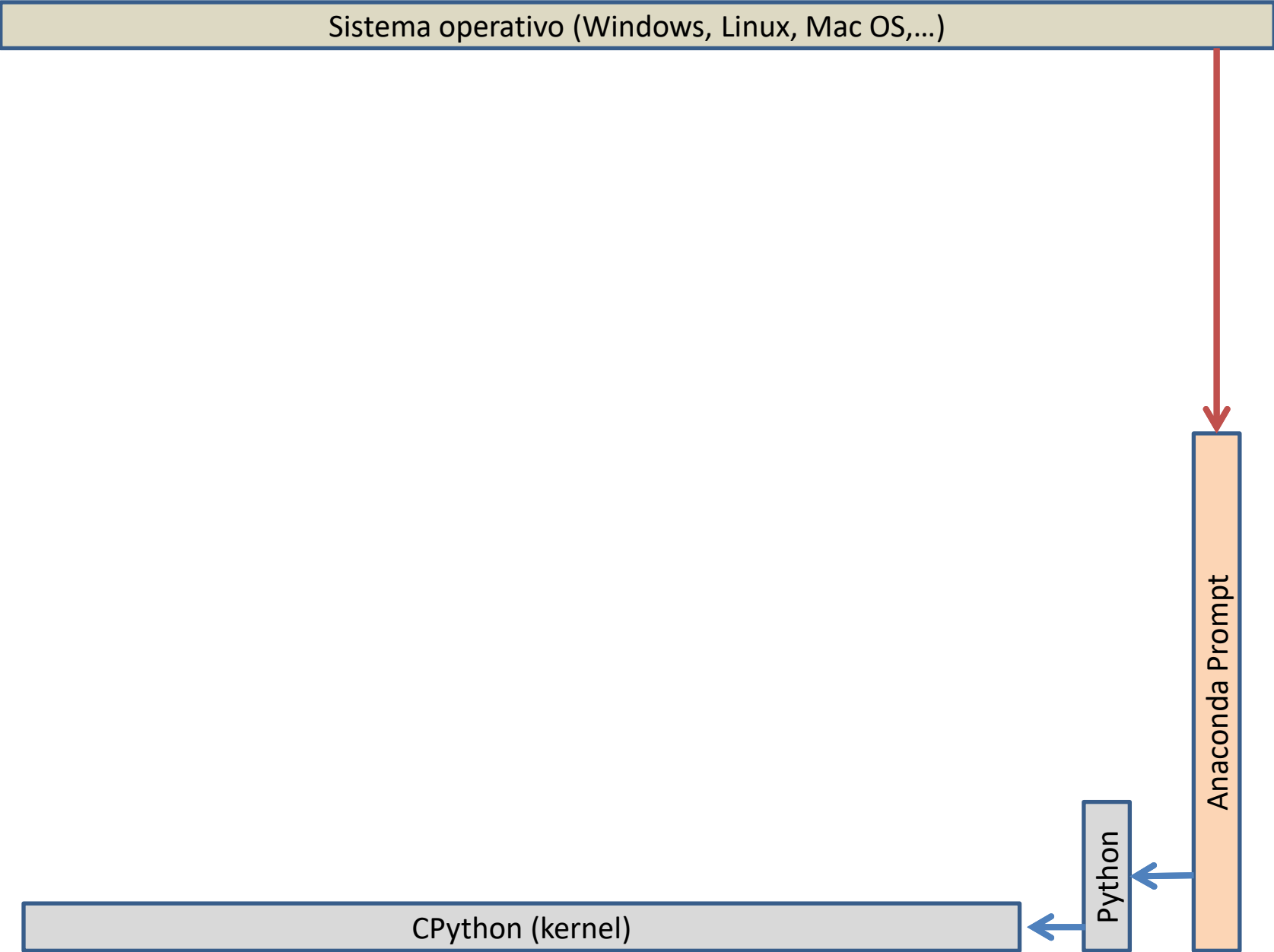
Linux



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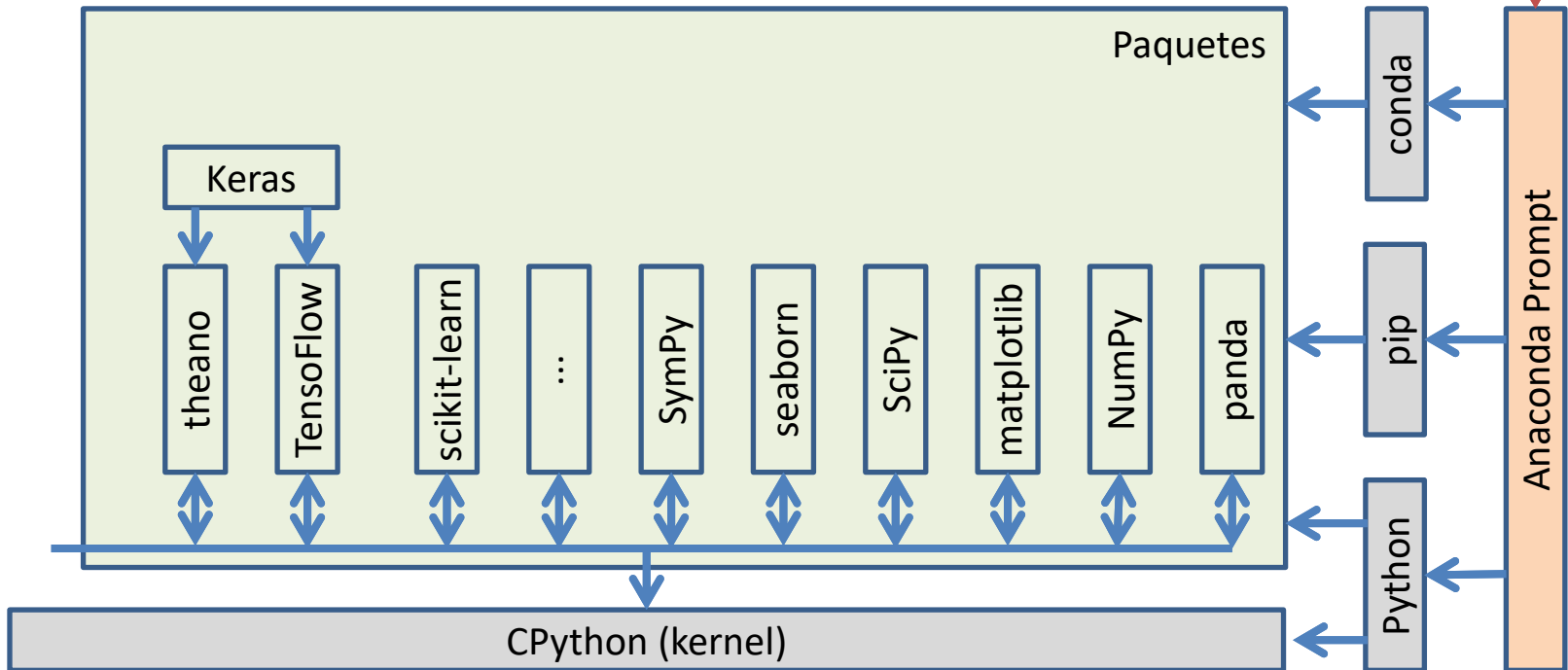


Anaconda Prompt · python



```
(base) C:\Users\Joaquin Luque>python
Python 3.7.3 (default, Mar 27 2019, 17:13:21) [MSC v.1915 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> a = 1
>>> b = 2
>>> c = a + b
>>> print(c)
3
>>>
```

Sistema operativo (Windows, Linux, Mac OS,...)



## Anaconda Prompt

```
(base) C:\Users\Joaquin Luque>pip show numpy
```

```
Name: numpy
```

```
Version: 1.16.2
```

```
Summary: NumPy is the fundamental package for array computing with Python.
```

```
Home-page: https://www.numpy.org
```

```
Author: Travis E. Oliphant et al.
```

```
Author-email: None
```

```
License: BSD
```

```
Location: c:\programdata\anaconda3\lib\site-packages
```

```
Requires:
```

```
Required-by: tensorflow, tensorboard, tables, seaborn, scikit-learn, PyWavelets,  
pytest-doctestplus, pytest-arraydiff, patsy, pandas, opt-einsum, numexpr, numba  
, mkl-random, mkl-fft, matplotlib, Keras-Preprocessing, Keras-Applications, h5py  
, EMD-signal, Bottleneck, bokeh, bkcharts, astropy
```

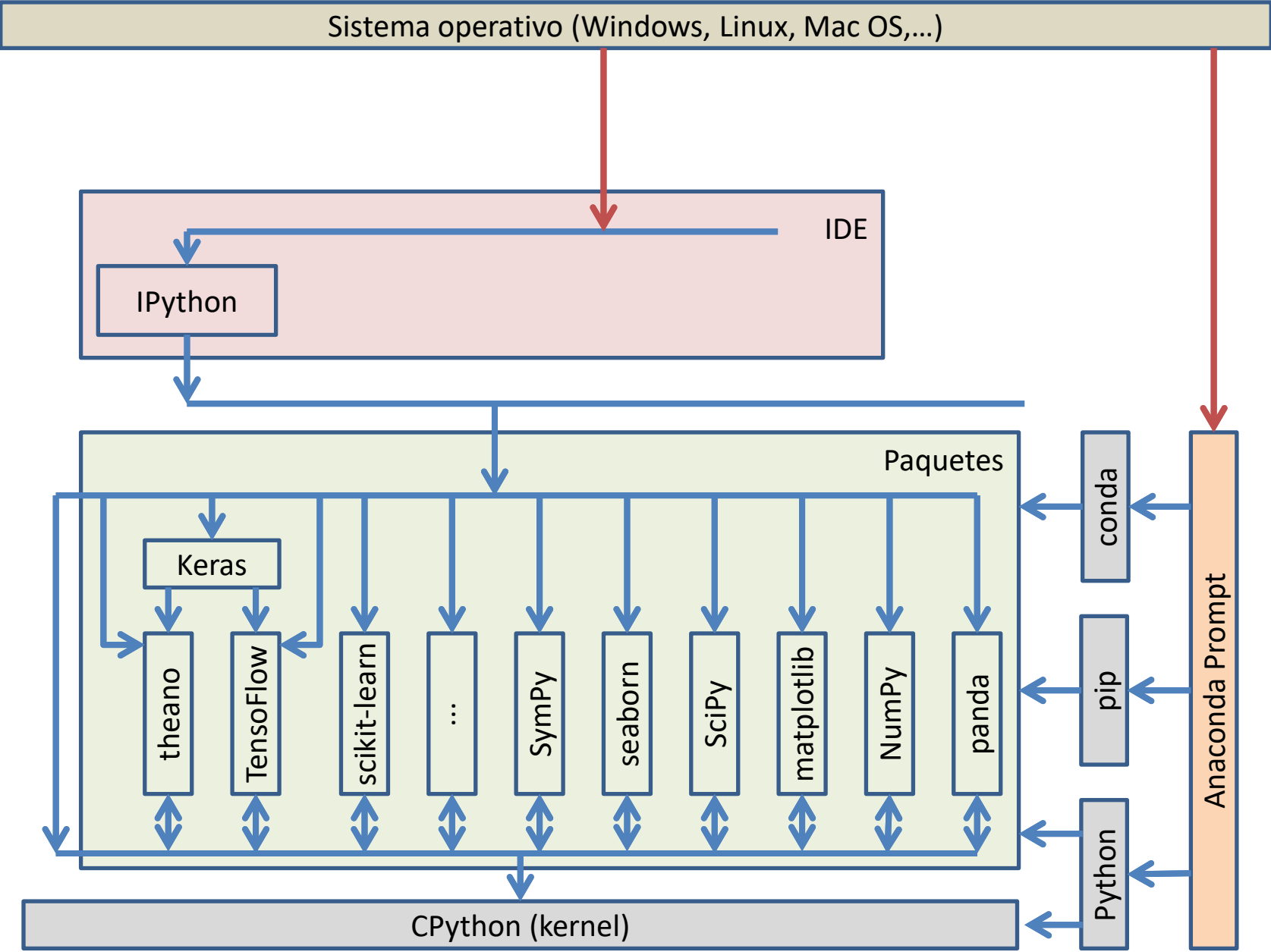
```
(base) C:\Users\Joaquin Luque>
```

# Anaconda Prompt

```
(base) C:\Users\Joaquin Luque>conda info
```

```
active environment : base
active env location : C:\ProgramData\Anaconda3
shell level : 1
user config file : C:\Users\Joaquin Luque\.condarc
populated config files : C:\Users\Joaquin Luque\.condarc
conda version : 4.6.11
conda-build version : 3.17.8
python version : 3.7.3.final.0
base environment : C:\ProgramData\Anaconda3 (read only)
channel URLs : https://repo.anaconda.com/pkgs/main/win-64
               https://repo.anaconda.com/pkgs/main/noarch
               https://repo.anaconda.com/pkgs/free/win-64
               https://repo.anaconda.com/pkgs/free/noarch
               https://repo.anaconda.com/pkgs/r/win-64
               https://repo.anaconda.com/pkgs/r/noarch
               https://repo.anaconda.com/pkgs/msys2/win-64
               https://repo.anaconda.com/pkgs/msys2/noarch
package cache : C:\ProgramData\Anaconda3\pkgs
                 C:\Users\Joaquin Luque\.conda\pkgs
                 C:\Users\Joaquin Luque\AppData\Local\conda\conda\pkgs
envs directories : C:\Users\Joaquin Luque\.conda\envs
                   C:\ProgramData\Anaconda3\envs
                   C:\Users\Joaquin Luque\AppData\Local\conda\conda\envs
platform : win-64
user-agent : conda/4.6.11 requests/2.21.0 CPython/3.7.3 Windows/10
Windows/10.0.18362
administrator : False
netrc file : None
offline mode : False
```

```
(base) C:\Users\Joaquin Luque>
```



Jupyter QtConsole 4.4.3

Python 3.7.3 (default, Mar 27 2019, 17:13:21) [MSC v.1915 64 bit (AMD64)]

Type 'copyright', 'credits' or 'license' for more information

IPython 7.4.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: a = 1

In [2]: b = 2

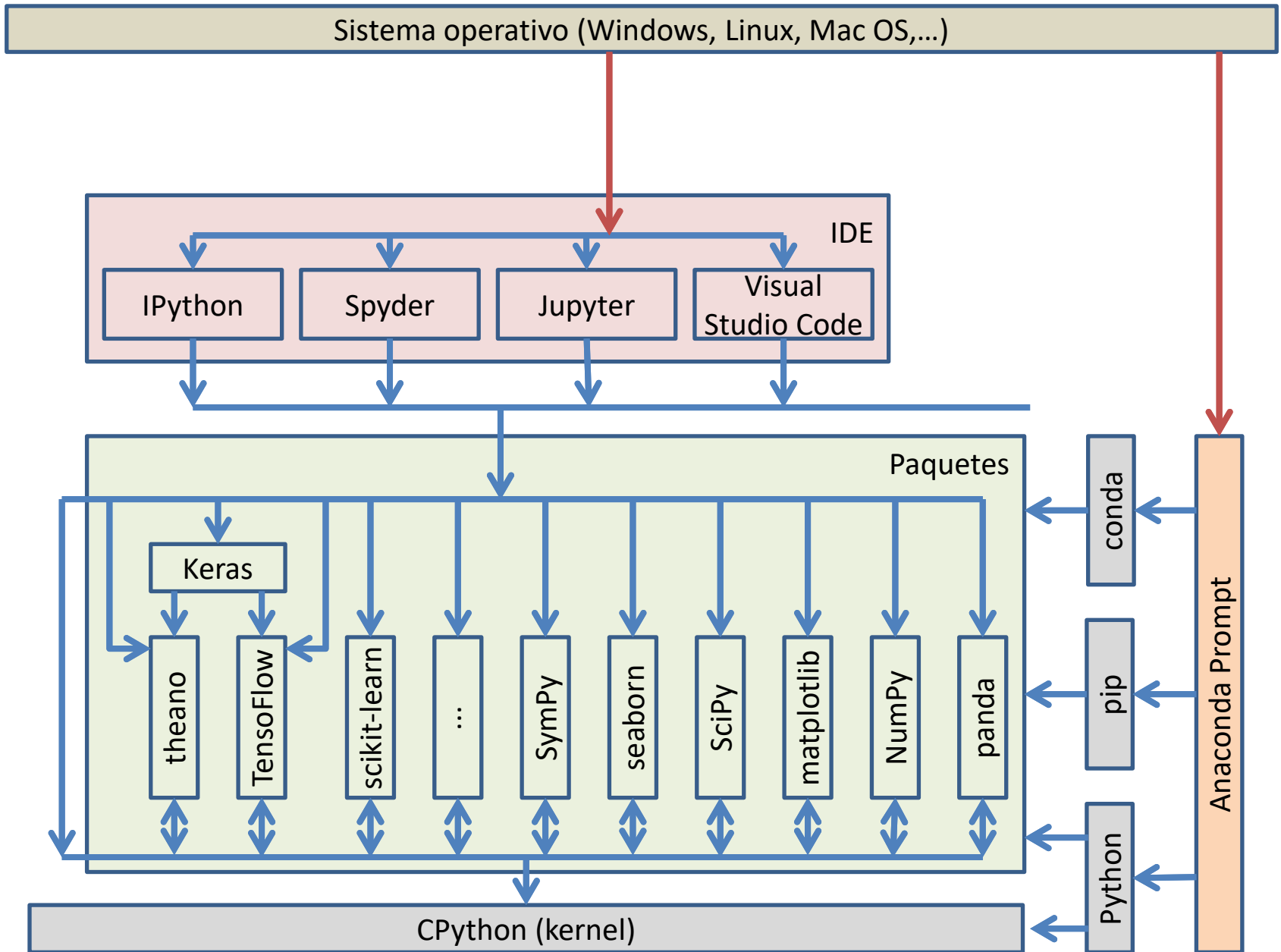
In [3]: c = a + b

In [4]: print(c)

3

In [5]: |





```
1 '''
2 Seminario: Introducción al Machine Learning
3
4 Ejercicio 1
5
6 Autor: Joaquín Luque
7 '''
8
9 #Import Libraries
10 import pandas as pd
11 import matplotlib.pyplot as plt
12
13
14 '''
15 Funciones
16 '''
17
18 def CargaDatos():
19     #x: vector de diseño (ingresos)
20     #y: vector de objetivos (target) (consumo)
21
22     df = pd.read_excel('DatosRegresion1D.xlsx')
23     x = df['ingresos']
24     y = df['consumo']
25     return x,y
26
27
28 def DibujaDatos(x,y,n):
29     #x: vector de diseño (ingresos)
30     #y: vector de objetivos (target) (consumo)
31     #n: nº de puntos a dibujar
32
33     plt.figure()
34     plt.plot(x[:n],y[:n], 'x')
35     plt.xlabel('Ingresos anuales (miles de €)')
36     plt.ylabel('Consumo anual de electricidad (Mw-h)')
37
38
39
40 '''
41 Cuerpo principal
42 '''
43
44 x,y = CargaDatos() #Apartado b
45 DibujaDatos(x,y,100) #Apartado c
46
47 plt.show()
48
```

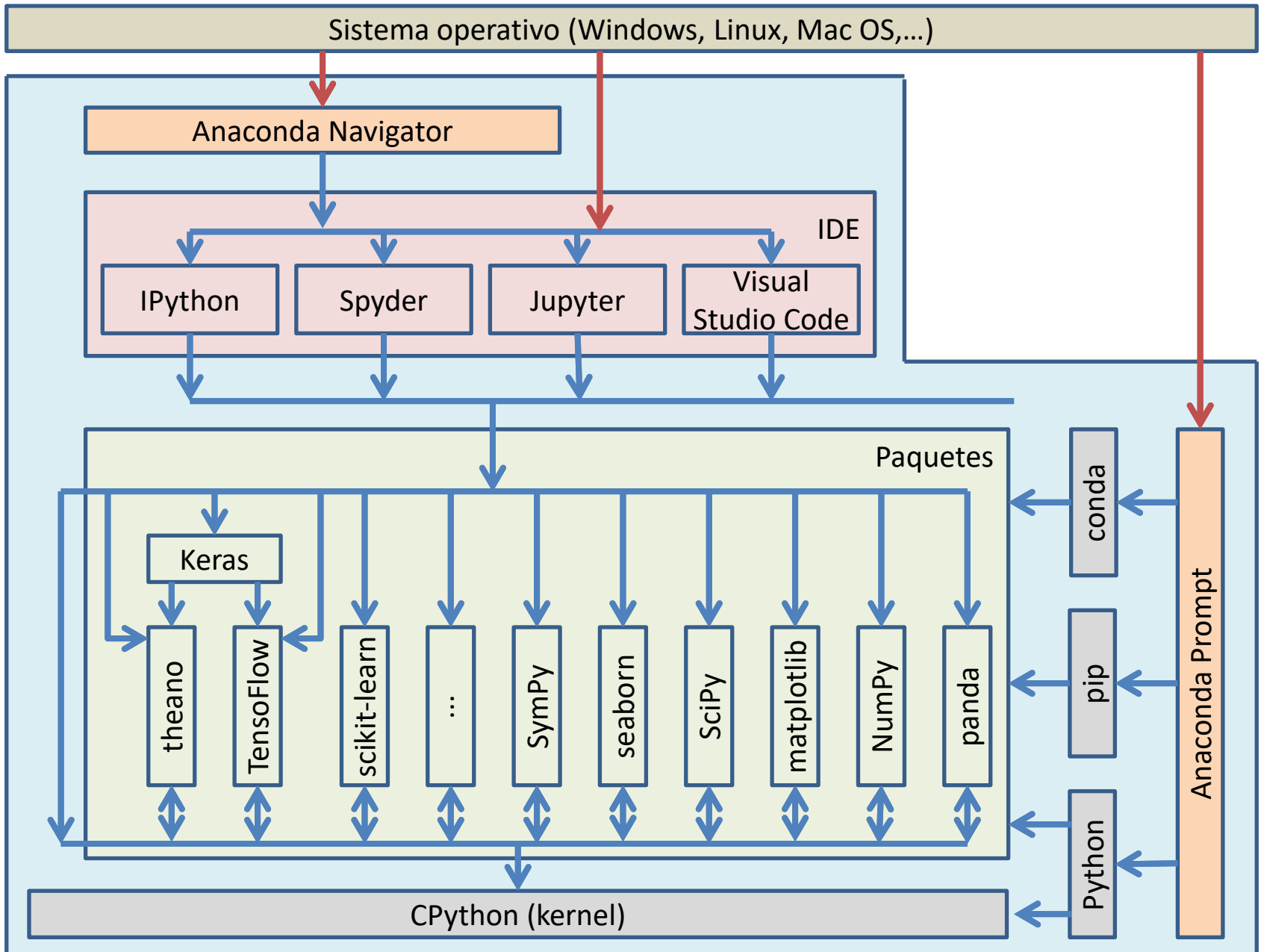
| Nombre | Tipo   | Tamaño  | Valor                                      |
|--------|--------|---------|--------------------------------------------|
| x      | Series | (1000,) | Series object of pandas.core.series module |
| y      | Series | (1000,) | Series object of pandas.core.series module |

Python 3.7.3 (default, Mar 27 2019, 17:13:21) [MSC v.1915 64 bit (AMD64)]  
Type "copyright", "credits" or "license" for more information.

IPython 7.4.0 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/Joaquín Luque/Documents/Articulos, Conferencias y Cursos/2019/Introduccion Machine Learning/Ejercicio\_1.py', wdir='C:/Users/Joaquín Luque/Documents/Articulos, Conferencias y Cursos/2019/Introduccion Machine Learning')

In [2]:



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JupyterLab

 0.35.4

An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.

Launch



Notebook

 5.7.8

Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.

Launch



Qt Console

 4.4.3

PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.

Launch

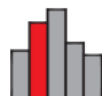


Spyder

 3.3.3

Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features

Launch



Glueviz

0.15.2

Multidimensional data visualization across files. Explore relationships within and among related datasets.

Install



Orange 3

3.23.0

Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.

Install



RStudio

1.1.456

A set of integrated tools designed to help



VS Code

1.39.0

Streamlined code editor with support for

